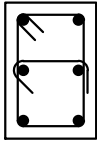
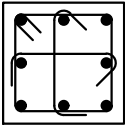


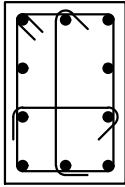
TYPE A
4 BARS (2 × 2)
TIE



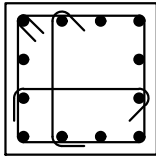
TYPE B
6 BARS (2 × 3)
TIE + 1 CROSSTIE



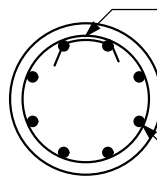
TYPE C
8 BARS (3 × 3)
TIE + 2 CROSSTIES



TYPE D
10 BARS (3 × 4)
TIE + 2 CROSSTIES



TYPE E
12 BARS (4 × 4)
TIE + 2 CROSSTIES



HOOP ENDS OVERLAP ≥ 150 mm,
135° HOOKS ROUND A BAR;
OVERLAPS OF SUCCESSIVE
HOOPS STAGGERED ROUND THE
COLUMN
 ≥ 6 BARS, EQUALLY SPACED

CIRCULAR HOOP

SCALE N.T.S.

2
1103

TABLE 12 - MAXIMUM TIE SPACING (mm)

BAR	ORD. RB9	DB10	DB12	INT. lo	SPEC. lo
DB16	255	255	255	125	95
DB20	320	320	320	160	120
DB25	–	400	400	200	150
DB28	–	445	445	220	150
DB32	–	–	510	255	150

TABLE 13 - COLUMN SCHEDULE CALL-UP (EXAMPLE)

MARK	SIZE (mm)	BARs	TYPE	TIES (@ mm)	FRAME
C1	400 × 400	8-DB20	C	RB9 @ 300	ORD.
C2	500 × 500	12-DB25	E	DB10 @ 200 (lo) / @ 400	INT.
C3	600 × 600	16-DB25	G	DB12 @ 125 (lo) / @ 150	SPE.

NOTES TO 1103

- THE COLUMN SCHEDULE CALLS UP A TIE TYPE BY ITS LETTER (TABLE 13). OTHER BAR COUNTS: THE SAME RULE - PERIMETER TIE, CROSSTIES ON ALTERNATE INTERMEDIATE BARS - SO THAT EVERY CORNER AND ALTERNATE BAR IS HELD AND NO BAR IS > 150 mm CLEAR FROM A HELD BAR. [EIT 011008 7.10.5.3]
- SPECIAL FRAMES: CROSSTIES OR HOOP LEGS AT $h_x \leq 350$ mm ACROSS THE SECTION (ADD CROSSTIES ON THE OTHER INTERMEDIATE BARS WHERE NEEDED). INTERMEDIATE AND SPECIAL FRAMES: DEFORMED HOOPS AND CROSSTIES, DB10 MIN.
- CROSSTIES: 135° + 6 db ≥ 75 mm AT ONE END, 90° + 6 db AT THE OTHER, ENGAGING THE PERIMETER BARS; THE 90° ENDS ALTERNATE BETWEEN SUCCESSIVE TIES AND BETWEEN OPPOSITE FACES. HOOKS OF SUCCESSIVE PERIMETER TIES AT DIFFERENT CORNERS.
- SPACING: TABLE 12, ALSO $\leq$ THE LEAST COLUMN SIZE (ORDINARY), $\leq c/2$ (INTERMEDIATE lo), $\leq c/4$ AND $s_0 = 100 + (350 - h_x)/3$, 100 – 150 (SPECIAL lo). OUTSIDE lo: INTERMEDIATE $\leq 2 s_0$, SPECIAL ≤ 6 db AND 150. ZONES: 1101.
- SPIRALS: CONTINUOUS BAR, SPACERS TO HOLD THE PITCH; RATIO PER DESIGN ($p_s \geq 0.45 (A_g/A_{ch} - 1) f_c/f_y$). [EIT 011008 7.10.4]
- SPECIAL FRAMES WHERE THE DESIGN APPLIES ACI 318-19 18.7.5.2(f) ($P_u > 0.3 A_g f_c$ OR $f_c > 70$ MPa): EVERY PERIMETER BAR HELD BY A HOOP CORNER OR A CROSSTIE WITH 135° HOOKS AT BOTH ENDS, $h_x \leq 200$ mm - TYPE EH. THE COLUMN SCHEDULE MARKS THESE COLUMNS; OTHERWISE NOTES 1 – 2 GOVERN (EIT 011008).

NOTES

- APPLIES TO ALL STRUCTURAL DRAWINGS.
- DIMENSIONS IN mm; LEVELS IN m.
- DO NOT SCALE FROM DRAWINGS.
- DRAFTING TO EIT 011006-19.

ISSUED FOR USE

KEY PLAN (N.T.S.)

NOT APPLICABLE
(TYPICAL DETAILS)

REVISIONS

REV	DESCRIPTION	DATE	SIGN
A	ISSUED FOR USE	03/10/2026	

OWNER / CLIENT APPROVAL

SIGNATURE :
NAME : DATE :

LOCAL AUTHORITY (BUILDING CONTROL)

PERMIT NO. :
[STAMP]

OWNER

[OWNER NAME]

PROJECT

[PROJECT NAME]
[BUILDING / LOCATION]

STRUCTURAL DESIGN OFFICE

[DESIGN OFFICE NAME]
[ADDRESS / TEL. / E-MAIL]

DRAWING TITLE

**TYPICAL COLUMN DETAILS (3)
TIE TYPES AND SPLICES**

DESIGNED :
DRAWN :
CHECKED :
ENGINEER :
COE LICENSE NO. :
SIGNATURE :

DRAWING NO.
STR-ST-1103-F-A
SHEET 3 OF 4 | A3
SCALE
N.T.S.
DATE
03/10/2026

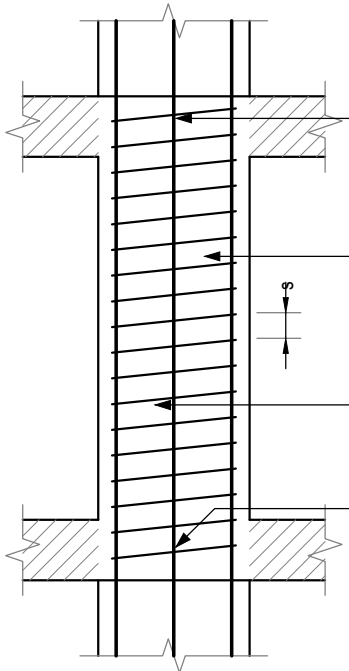
APPROVED BY : DATE :

TIE TYPES

SCALE N.T.S.

1
1103

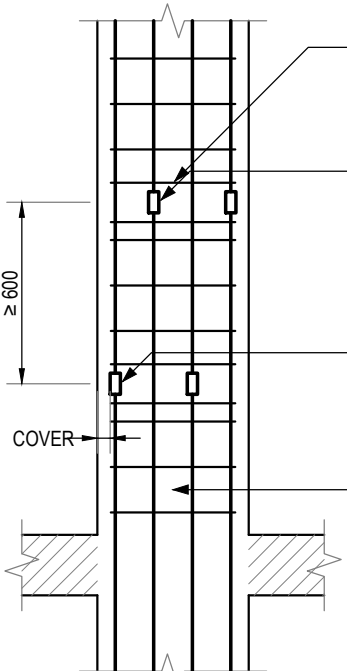
SECTION SIZES ILLUSTRATIVE; THE TYPE IS SET BY THE BARS PER FACE



SPIRAL COLUMN

SCALE N.T.S.

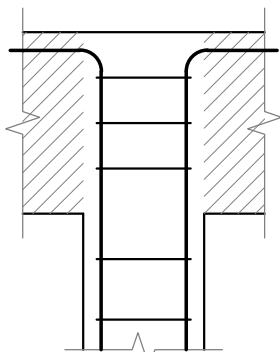
3
1103



MECHANICAL SPLICES

SCALE N.T.S.

4
1103



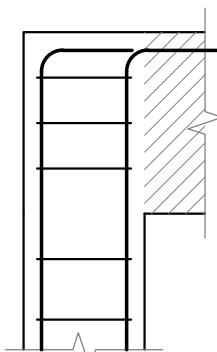
(a) INTERIOR: HOOKS OUTWARD

COLUMN TOP AT ROOF

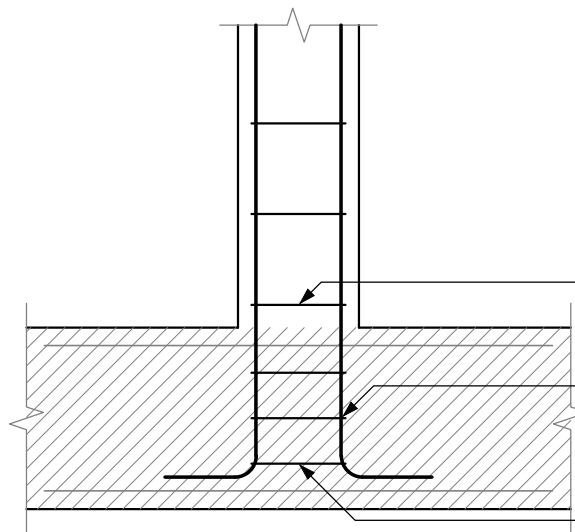
SCALE N.T.S.

90° HOOK 12 db AT THE ROOF TOP AND ≥ l_{dh} ABOVE THE SOFFIT; TIES TO THE TOP

1
1104



(b) EDGE / CORNER: HOOKS INWARD



COLUMN ON A TRANSFER BEAM

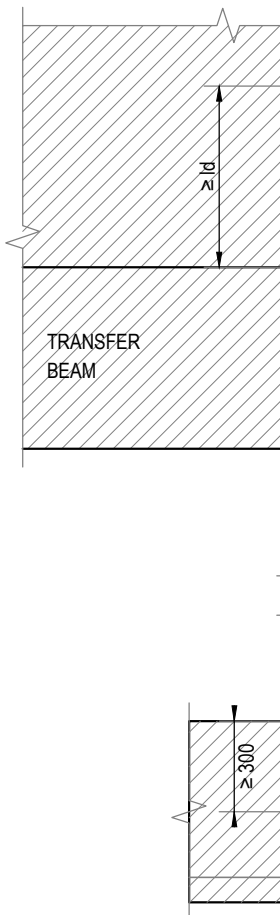
SCALE N.T.S.

2
1104

FIRST TIE ≤ s/2 ABOVE THE BEAM

COLUMN BARS TO THE BEAM
BOTTOM LAYER, 90° HOOKS
OUTWARD ON THE BOTTOM BARS

COLUMN TIES CONTINUE INTO THE
BEAM AT THE SAME SPACING



COLUMN UNDER A DISCONTINUED WALL

SCALE N.T.S.

SPECIAL FRAMES, $P_u > A_g f_c / 10$

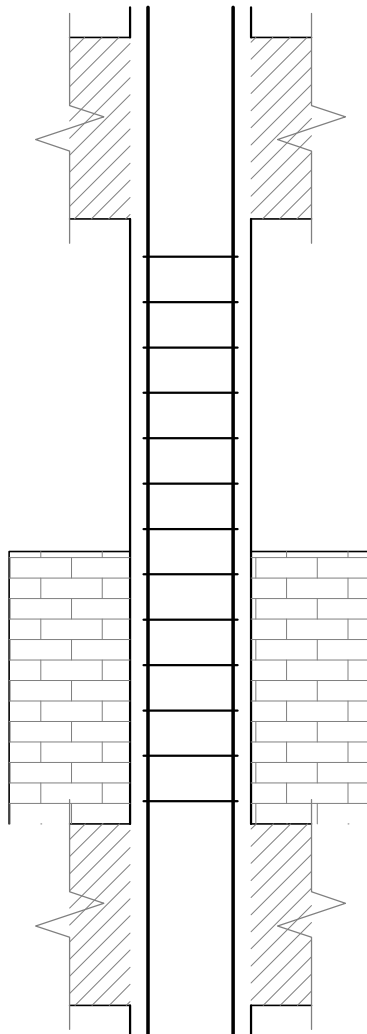
3
1104

WALL ABOVE, DISCONTINUED BELOW
COLUMN BARS ≥ l_d INTO THE WALL

HOOPS THROUGH THE BEAM, ≥ l_d
(LARGEST COLUMN BAR) INTO THE
WALL

HOOPS @ s₀ OVER THE FULL HEIGHT OF
EVERY STOREY BELOW THE WALL

HOOPS ≥ 300 mm INTO THE FOOTING
(ON A WALL: ≥ l_d)



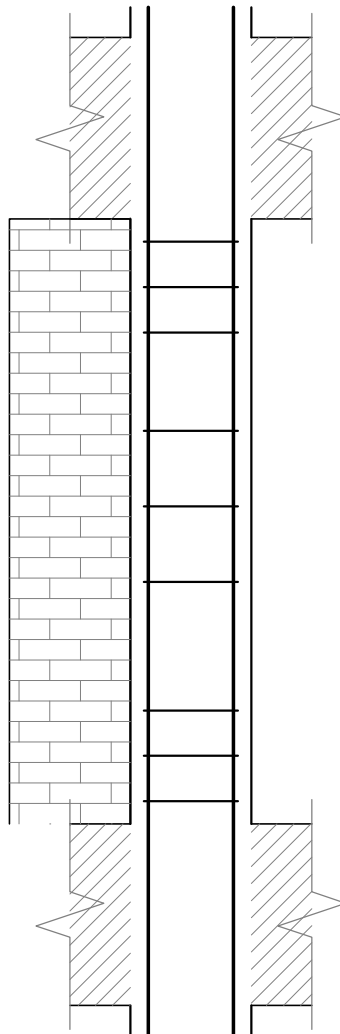
(a) PARTIAL-HEIGHT INFILL
HOOPS @ s₀ OVER THE FULL HEIGHT;
SHEAR FOR THE SHORT COLUMN;
HOOPS ≥ c₁ INTO THE WALL ZONE

COLUMN BESIDE MASONRY INFILL

SCALE N.T.S.

DPT 1301/1302 5.2.5.1 (FRAMES 2, 3)

4
1104



(b) FULL INFILL ON ONE SIDE
l_o ZONES @ s₀;
MID ZONE ≤ 2 s₀ AND ≤ d/2

NOTES

1. APPLIES TO ALL STRUCTURAL DRAWINGS.
2. DIMENSIONS IN mm; LEVELS IN m.
3. DO NOT SCALE FROM DRAWINGS.
4. DRAFTING TO EIT 011006-19.

ISSUED FOR USE

KEY PLAN (N.T.S.)

NOT APPLICABLE
(TYPICAL DETAILS)

REVISIONS

REV	DESCRIPTION	DATE	SIGN
A	ISSUED FOR USE	03/10/2026	

OWNER / CLIENT APPROVAL

SIGNATURE :

NAME : DATE :

LOCAL AUTHORITY (BUILDING CONTROL)

PERMIT NO. :

[STAMP]

OWNER

[OWNER NAME]

PROJECT

[PROJECT NAME]

[BUILDING / LOCATION]

STRUCTURAL DESIGN OFFICE

[DESIGN OFFICE NAME]

[ADDRESS / TEL. / E-MAIL]

DRAWING TITLE

**TYPICAL COLUMN DETAILS (4)
COLUMN ENDS, SPECIAL CASES**

DESIGNED :

DRAWN :

CHECKED :

ENGINEER :

COE LICENSE NO. :

SIGNATURE :

DRAWING NO.

STR-ST-1104-F-A

SHEET 4 OF 4 | A3

SCALE

N.T.S.

DATE

03/10/2026

APPROVED BY : DATE :